

Fleet Route Optimization Report

Generated (UTC): 2026-08-29T15:11:43.034322+00:00 | Use Case: GENERIC
Algorithm: QPSO-VRP | Data Source: FALLBACK TRAFFIC

1. Executive Summary

Total Fleet Distance:	909.2 km
Total Travel Time:	646.2 min (10.77 hrs)
Time Saved (Optimization):	5.7 hrs vs baseline
CO2 Reduction Offset:	27.3 kg CO2 reduction
Estimated Fuel & Cost:	Rs.7274.0 (~75.8 L fuel)
Traffic Delay Estimate:	N/A (fallback mode)
On-Time Delivery Rate:	0.0%
Vehicles / Customer Stops:	1 vehicles / 1 stops

2. Vehicle Route Manifests

Vehicle #1 - Distance: 909.2 km | Duration: 646.2 min (10.77 hrs)

Seq	Stop Name	Arrival (h)	Leg Dist (km)	Traffic	Status
1	Rourkela	8.0h	0.0	UNKNOWN	-
2	Kanpur	24.43h	821.75	UNKNOWN	LATE

3. Traffic & Congestion Analysis

Segment Breakdown: High Congestion: 0 | Medium: 0 | Low: 0
Note: Fallback free-flow traffic model used. Figures assume ideal road conditions.

4. Optimization Performance

Total Iterations: 100 | Initial Cost: 20290.8 | Final Optimal Cost: 20290.8
Convergence Rate: None | Quantum Tunneling Events: 0 | Runtime: 2.8436s

5. Recommendations & Operational Notes

- Total distance, time, and cost figures above summarize this specific optimization run and are based on the data source noted in metadata.
- Compare the optimization_performance section across repeated runs to gauge consistency of the solver on your data.
- Data source for this report: fallback traffic data, as noted in metadata.data_source. Fallback-mode figures assume free-flow conditions and will understate real-world delay.